The exploration, explained from the paper outward

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Written in reply to the note of 13 August. Companion to `reports/progress_report.tex`; every number here is reproduced by the scripts in `reports/probe/`.

Introduction

The note of 13 August asked two questions of my 12 August log entries: what the key discrepancy is, and what directions follow from it. Those entries state their conclusions without building up to them, which is fair criticism. This document is the chain that leads to them, starting from the definitions in the papers themselves. This introduction gives the whole argument in outline — the three papers it rests on, what appears to be wrong and why, how each of those things was checked, and what came out — and every claim in it is built up from first principles later on.

The three papers

Three papers were replicated in full before any of this began; they were treated as a literature review rather than as ends in themselves. Two of them supply the crisis indicators that this exploration compares.

Paper	What it builds	Its indicator	How the rebuild came out
Paper 1. Cardoso & Palomar (2020), *Learning Undirected Graphs in Financial Markets*	Learns a graph Laplacian $\mathbf{L}$ *directly* from returns by penalised maximum likelihood under a Gaussian Markov random field — no correlation table, no threshold.	$\lambda_2(\mathbf{L})$, the **Fiedler value**: how tightly the graph holds together. Small means close to falling apart.	The method, the monthly path of λ_2 and the COVID exit all reproduce. The trading result does not: the strategy ends at -0.011 where buy-and-hold ends at $+0.078$, with no threshold moved to rescue it.
Paper 2. Hallac, Vare, Boyd & Leskovec (2017), *TICC*	Cuts a multivariate series into stretches, each with its own block-Toeplitz dependence pattern, so a regime can recur later. Not a network method.	None — it labels regimes rather than scoring days.	Headline accuracy reproduces; the ablations come out weaker than reported, so the rebuild makes the method look *more* necessary than the paper does. Peripheral here, but the only one modelling a state that persists in time.
Paper 3. Silva et al. (2015), *Modular Dynamics of Financial Market Networks*	One network per day: keep the strongest 10% of pairwise correlations as edges, run Louvain, record how cleanly the stocks split into groups. 6,315 daily networks, 360 stocks, 25 years.	$Q(t)$, **modularity**: a single number per day for how group-like the market is.	Reproduces on its own terms. What the number *means* is the problem, and that is this document's subject. A separate 26-item audit of it is already on file.

The prompt behind this work (§1) was to put Paper 1's indicator and Paper 3's side by side, and to look at the **modularity matrix** — the matrix Q is computed from — rather than only at Q itself. Papers 1 and 3 therefore supply the two candidates; Paper 2 sits to one side as a possible comparator.

What appears to be wrong, and what would explain it

Five things came out of the exploration that do not fit the papers' own account of themselves. The middle column is a candidate explanation, not a settled one; the last says where the reasoning is set out in full.

What was observed	What would explain it	Where
$Q(t)$ barely separates crisis days from calm ones (AUC 0.538, coin flip 0.500), while three other readings of *the same* matrix reach 0.722 to 0.760.	The reduction, not the matrix. Q collapses 129,600 matrix entries to about nine groups to one number; almost everything is discarded on the way. The discarded threshold $\tau(t)$ alone scores 0.807.	§5
Louvain reports a median of *nine* communities every day, yet i.i.d. Gaussian noise pushed through the identical pipeline scores $Q = 0.2246$ against the real market's 0.2215.	Thresholding a correlation matrix yields a triangle-rich, transitive graph by construction, and such graphs always look modular. The partition may be an artifact of the construction rather than a property of the market.	§6.2
At a 30-day window on 360 stocks, the average number of correlation eigenvalues escaping the random-matrix noise band is **0.70**, and zero on 2,668 of 6,315 days.	The window is too short for the question. Thirty observations give a correlation matrix of rank at most 29, so most of what is called community structure is not statistically resolvable at the window length this literature itself uses.	§6.3
Theory makes the two indicators one number — $\lambda_2(\mathbf{L}) + \mu_1(\mathbf{B}) = d$ on a d-regular graph, and Silva pins the mean degree at 35.90 daily — but the measured sum is 26.41, and the two correlate at -0.187 where the identity demands -1.	The identity's premises fail in three separable ways: degree heterogeneity (fixed *average* degree is not a regular graph), fragmentation (these networks are disconnected on 63% of days), and the market mode. Which of the three carries the crisis information is an open, answerable question.	§7
The three highest-scoring detectors correlate above 0.99 with each other, and every variable added to them helps in-sample while hurting out-of-sample on a strict pre/post-2002 split.	They are one signal, not three: the market mode, published as the absorption ratio in 1999. The in/out-of-sample gap is the textbook overfitting signature, though 27 test events is too few to settle it.	§8

How each of those was checked

Nothing above rests on a new pipeline. Silva's replication already builds one network per day for 6,315 days, and each check reuses it unchanged so that old and new numbers are directly comparable.

- **One ruler for everything** (§4). Every candidate is turned into a 250-day trailing z-score computed on strictly prior data, then scored by AUC against crisis windows fixed in advance (17 events, 2.68% base rate). As a control the probe re-derives Silva's own stored Q and τ scores to ten decimal places, so the new numbers sit on exactly the ruler the old ones do.
- **Full spectra rather than one statistic.** All eigenvalues of three operators — the Laplacian $\mathbf{L}$, the modularity matrix $\mathbf{B}$ and the correlation matrix $\mathbf{C}$ — on each of the 6,315 networks, with every resulting statistic scored on that same ruler. This is what makes the matrix-versus-scalar comparison a measurement rather than an argument.
- **The identity tested directly**, and then re-tested on the largest connected component only, which removes fragmentation as a confound and lets its contribution be separated from the rest.
- **Matched nulls taken from the replication itself**: i.i.d. Gaussian returns through the identical construction, and Louvain re-run across ten seeds to measure how much of the daily variation in Q is the algorithm rather than the market.

- **Incremental value, in-sample and out.** Whether the modularity spectrum adds anything to the market mode, judged both in-sample and on a strict chronological split at 2002.
- **Arithmetic re-derived.** A checking script recomputes all 91 values quoted in this document from the stored artifacts and confirms each printed string appears in the source, so the prose cannot drift from the data in either direction (Appendix A).

Summary

The discrepancy, in one sentence.

Silva et al. reduce a 360×360 matrix to a single number $Q(t)$; that number is almost uninformative about crises (AUC 0.538, against 0.500 for a coin), while three other readings of *the same matrix, on the same networks, under the same test* score 0.722 to 0.760 — so the information was present, and the reduction is where it was lost. (§5)

That is the answer to the first question. Around it sit a second, structural discrepancy — the two candidate indicators are provably the same number and empirically unrelated (§7) — and one finding that cuts against the whole exercise: at the window length in use there may be very little community structure to detect in the first place (§6.3). The strongest detectors, meanwhile, are one already-known signal, and nothing yet beats it out of sample (§8).

On the second question, three directions follow. They are set out in §9 in a common frame, at equal length, with no ordering implied and no combined recommendation.

- **D1** — decompose the 9.49 residual into its degree, fragmentation and market-mode parts, and compare the Laplacian and modularity families as one structural question rather than as a leaderboard.
- **D2** — map where on the (window length, universe size, density) grid community detection in correlation networks is statistically possible at all, and replace the Louvain community count with a deterministic one.
- **D3** — the negative end of the modularity spectrum: anti-communities, groups trading against each other, which thresholding structurally cannot see.

Appendix A lists where every number quoted above comes from and how to re-derive it.

1 Where the question came from

The question was not what those three papers claim — that is settled in the replications — but what to build on top of them. That question came from my supervisor:

> *Paper 1's crash indicator is the Fiedler value $\lambda_2(\mathbf{L})$. Paper 3's is modularity Q. Look at the **modularity matrix** as well — its eigenvalues and singular values may carry more. Can we get a study out of comparing Laplacian against modularity matrices?*

Three things were done in response.

The repository was re-read end to end — the three pipelines summarised in the introduction, together with the 26-item methodological audit already run against Silva.

The outside literature was searched on two fronts: spectral market-stress indicators in finance, and the mathematics of the modularity matrix. The mathematics lives almost entirely outside finance — MacMahon & Garlaschelli on the correct random-matrix null for correlation-based modularity, Fasino & Tudisco on the modularity spectrum and anti-communities, Nadakuditi & Newman on when community detection is possible at all. In finance, modularity appears essentially

only as the scalar Q. Repeated searches for modularity-matrix *spectra* used as financial indicators returned nothing.

How far this search actually went.
Not far, and I want to be straight about it. This survey was done largely *with* Claude rather than by reading the papers through: I used it to find the relevant work, to summarise what each result says, and to check that no existing paper already does what §9 proposes. I have read abstracts, the statements of the main results, and enough of MacMahon & Garlaschelli and Fasino & Tudisco to use their results here — but I have not worked through the proofs, and I have not audited the finance side for papers the searches missed. So treat the claim "nobody in finance has used the modularity *spectrum*" as a well-supported absence of evidence, not as a settled literature review. It needs a proper one before it is put in writing anywhere that matters.

Rather than speculate about what the modularity matrix would show, it was computed. Silva's pipeline already builds one network per day for 6,315 days. The probe takes the full spectrum of three operators on each of those networks and scores every resulting statistic through the replication's *own* crisis detector — reproducing the replication's stored scores to ten decimal places as a control, so the new numbers sit on exactly the ruler the old ones do.

The rest of this document is that chain, in order: what Silva computes (§2), what matrix sits underneath it (§3), how anything gets scored (§4), what came out (§5–§8), and what could be done next (§9).

2 What Silva et al. actually compute

Everything below is the replication's own code path (`net/construct.py` in the Paper-3 tree), at full scale: 360 NYSE common stocks, January 1986 to February 2011.

Step 1 — returns. Daily log returns on adjusted closes, $Y_i(t) = \ln P_i(t) - \ln P_i(t-1)$. A $360 \times 6{,}344$ panel.

Step 2 — a correlation matrix per day. Slide a $\Delta t = 30$ trading-day window forward one day at a time. In each window, standardise each stock's returns and form the 360×360 Pearson correlation matrix $\mathbf{C}(t)$. That is $\binom{360}{2} = \mathbf{64{,}620}$ pairwise correlations, each a real number in $[-1, 1]$. Sliding to the end of the panel gives **6,315** of these.

Step 3 — threshold to a graph. Keep only the strongest correlations. Crucially the threshold is *not* fixed: each day $\tau(t)$ is re-solved so that exactly 10% of pairs survive,

$$A_{ij}(t) = \Theta(\rho_{ij}(t) - \tau(t)) - \delta_{ij}, \qquad \#\{\text{edges}\} = 0.10 \cdot \tbinom{360}{2} = \mathbf{6{,}462} \ \text{ every day.}$$

Two consequences worth holding onto. First, $2m = 12{,}924$ and the average degree is $12{,}924/360 = \mathbf{35.90}$ — *exactly*, every day, by construction, not approximately. Second, $\tau(t)$ itself is a real quantity that varies a lot (mean 0.444, max 0.861 on Black Monday); the pipeline computes it, uses it, and discards it.

Step 4 — communities. Run Louvain on the resulting graph. It returns a partition of the 360 stocks into groups — a median of 9 of them, though on 1987-10-19 it returns 156, of which 152 are single stocks.

Step 5 — one number. Score that partition with Newman's modularity,

$$Q = \sum_c \left[\frac{e_c}{m} - \left(\frac{d_c}{2m} \right)^2 \right],$$

and record it. $Q(t)$ is the paper's central quantity; its 25-year trajectory is the paper's story.

Figure 1 draws that chain with what each step throws away. Nothing in it is a criticism yet — every pipeline discards something, and a scalar indicator is the point of the exercise. The question

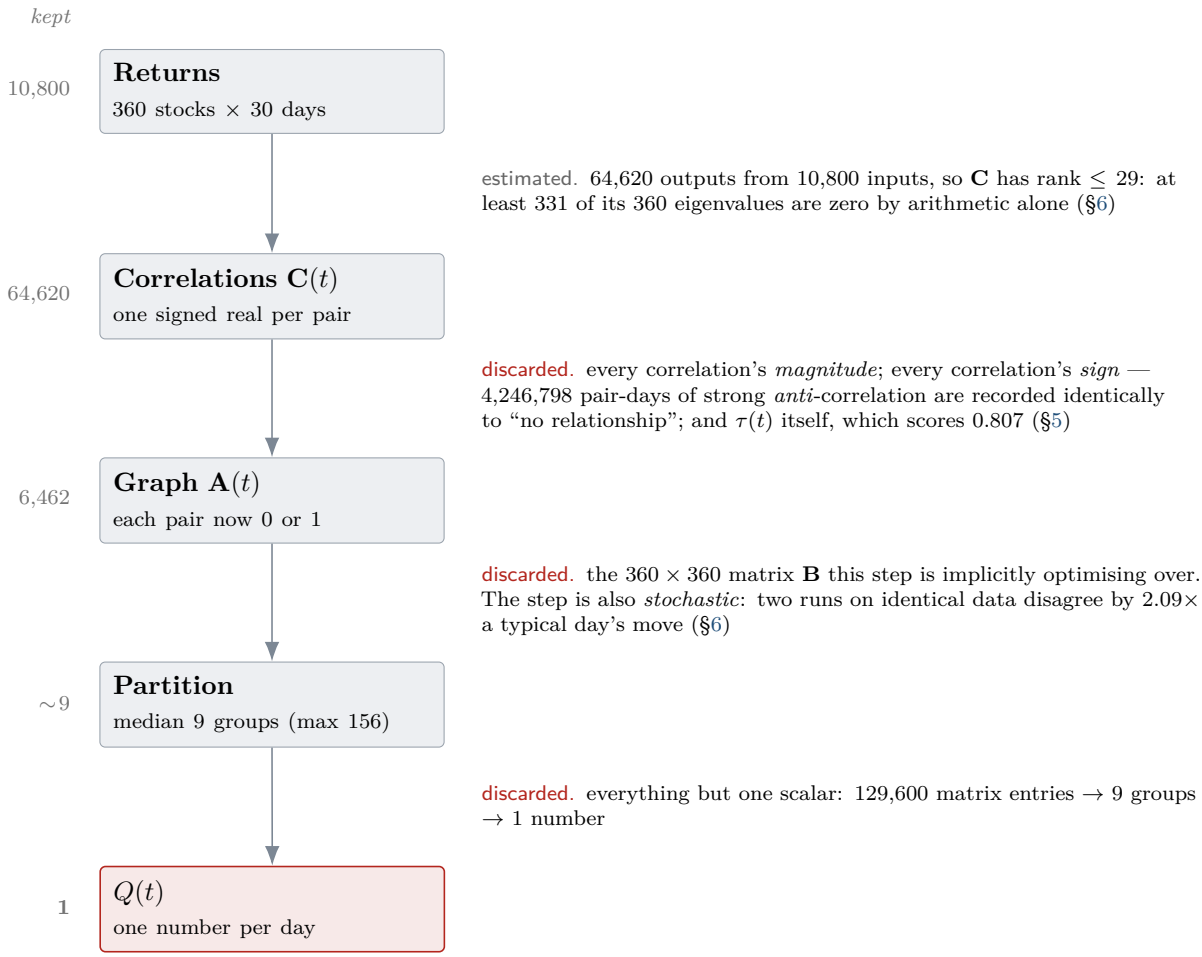


Figure 1: Silva's pipeline as a sequence of reductions; the right-hand column is what each arrow costs. The paper reports the bottom box. The exploration in this document consists entirely of asking what is in the boxes above it, and scoring those contents on the paper's own ruler.

§5 answers is whether *this particular* discard was expensive, and it is answerable because the discarded objects can simply be scored too.

3 The matrix underneath Q

Q is not a primitive quantity. It is a summary of a specific matrix, and that matrix is the object the prompt was about. This section builds it from scratch.

3.1 The question modularity is really asking

Suppose stocks i and j have an edge. Is that interesting? Not necessarily. If i is connected to 200 of the other 359 stocks, an edge to j says almost nothing — i is connected to nearly everyone. But if i and j each have only five edges and one of them is to each other, that is meaningful.

So a baseline is needed: *how many edges would we expect between i and j purely by accident, given how many edges each of them has?* Writing d_i for the degree of stock i (its number of edges) and $2m$ for the total number of edge endpoints, the standard answer — the configuration model — is

$$\mathbb{E}\left[A_{ij}\right] \;=\; \frac{d_i d_j}{2m}.$$

The intuition is a matching argument: stock i has d_i endpoints dangling, each picks a partner at

random from the $2m$ endpoints in the network, and d_j of those belong to j.

Check it on the real numbers, where $2m = 12{,}924$. Two typical stocks, each of degree 36:

$$\frac{36 \times 36}{12{,}924} = 0.100,$$

a 10% chance — exactly the density of the network, as it must be. Now two hubs, each of degree 100:

$$\frac{100 \times 100}{12{,}924} = 0.774.$$

Those two are expected to be connected almost certainly, just from being popular. An edge between them is not news.

3.2 The modularity matrix

The **modularity matrix** is what was observed minus what chance predicts:

$$\mathbf{B} \;=\; \mathbf{A} - \frac{\mathbf{d}\mathbf{d}^{\top}}{2m}, \qquad B_{ij} = A_{ij} - \frac{d_i d_j}{2m}.$$

It is the same size as $\mathbf{A}$, 360×360, and every entry is a surprise score. Table 1 reads off the four cases using the real numbers just computed.

Table 1: Entries of $\mathbf{B}$ as surprise scores, using the two expectations computed above. The last row is the one to keep hold of.

observed	expected	B_{ij}	reading
edge (1)	0.10	$+0.90$	strong surprise: together far more than chance
edge (1)	0.77	$+0.23$	two hubs linked; barely news
no edge (0)	0.10	-0.10	ordinary; most pairs look like this
no edge (0)	0.77	$\mathbf{-0.77}$	two hubs *avoiding* each other

That last row is the structural point. $\mathbf{B}$ has a meaningful negative end: it can say *"an edge should be here and is not."* The adjacency matrix is 0/1 and the Laplacian is built from $\mathbf{A}$ and the degrees, so **neither can express that**. The asymmetry is real, and it is where direction D3 comes from.

Figure 2 shows the subtraction on a small graph where the arithmetic can be checked by eye.

3.3 Where Q comes from

Newman's Q is now easy to state. Pick a way of splitting the stocks into groups; add up $\mathbf{B}$ over all pairs that landed in the *same* group; divide by $2m$:

$$Q \;=\; \frac{1}{2m} \sum_{i,j \text{ in the same group}} B_{ij}.$$

That is all it is. Q **is a sum of selected entries of** $\mathbf{B}$, and Louvain is a greedy randomised search for the grouping that maximises that sum. So the chain of Figure 1 ends:

$$\underbrace{\mathbf{B},\ 129{,}600 \text{ entries}}_{\text{a } 360\times360 \text{ matrix}} \longrightarrow \underbrace{\text{one grouping}}_{\text{stochastic search}} \longrightarrow \underbrace{Q}_{\text{one number}}.$$

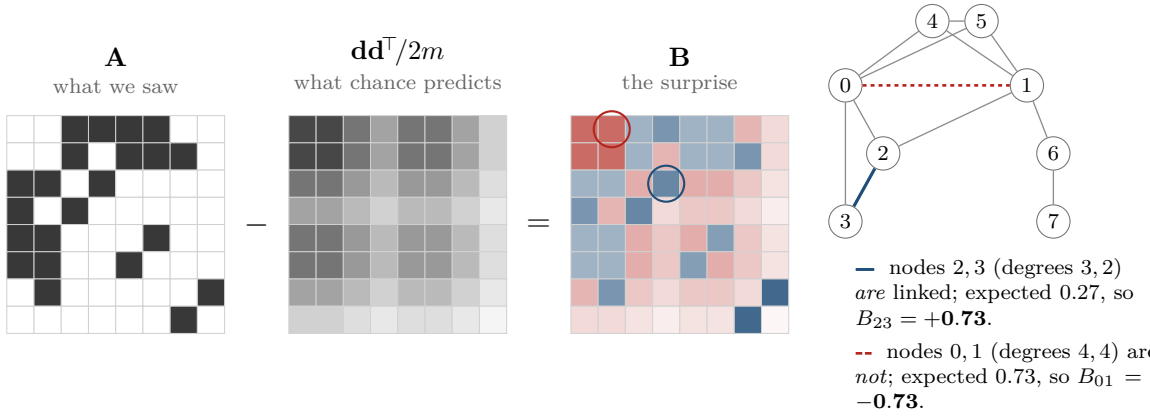


Figure 2: $\mathbf{B} = \mathbf{A} - \mathbf{dd}^\top/2m$ on an eight-node graph, drawn to scale; blue is positive surprise, red negative, intensity $\propto |B_{ij}|$. The two circled cells have *identical magnitude and opposite sign* — yet only one of them corresponds to an edge that exists. Subtracting the chance baseline turns a 0/1 matrix into a signed one, and what appears is exactly the information about pairs that are *less* connected than they should be. Every row of $\mathbf{B}$ sums to zero.

3.4 Reading the matrix directly

An **eigenvector** of a matrix is a direction the matrix does not rotate, only stretches or shrinks; the **eigenvalue** is the stretch factor. A symmetric 360×360 matrix has 360 such pairs and they describe it completely. The list of eigenvalues is its **spectrum**.

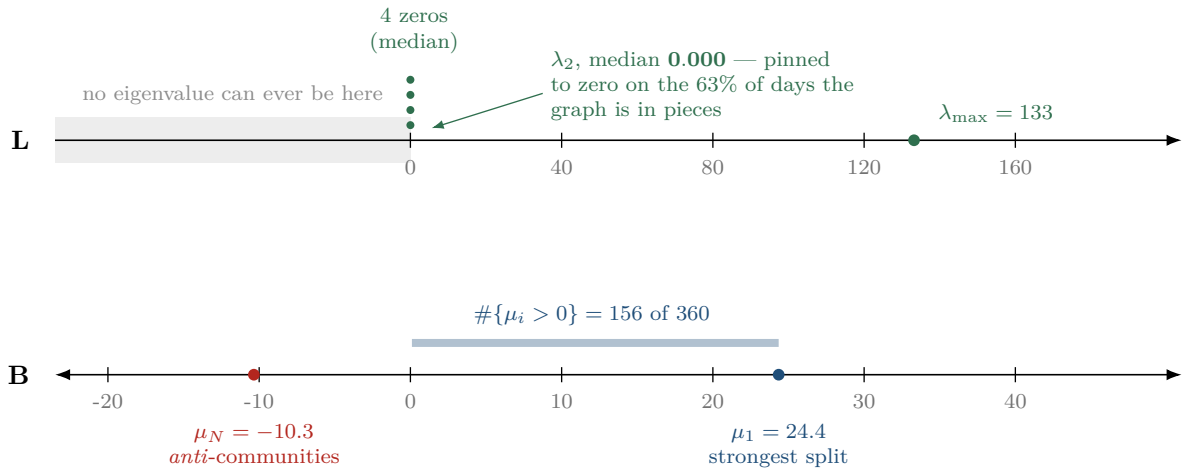
Two structural facts about $\mathbf{B}$, both used later:

- **Every row sums to zero.** Node i's observed edges sum to d_i, and its expected edges sum to $d_i \left(\sum_j d_j\right)/2m = d_i$. They cancel exactly, so $\mathbf{B}$ always has a zero eigenvalue.
- **$\mathbf{B}$ is indefinite** — it always has both positive and negative eigenvalues. Its diagonal is entirely negative, so its trace is negative, forcing negative eigenvalues; and the zero row sums force positive ones. This is the sharp contrast with the Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$, which is positive semi-definite: every eigenvalue of $\mathbf{L}$ is ≥ 0.

Writing $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_N$ for the modularity spectrum, four quantities are available without running any algorithm:

- **μ_1, the largest.** Its eigenvector is the single strongest "these stocks against those" split in the surprise matrix. Newman's original method reads the signs of this eigenvector to bisect the network. So μ_1 is doing Q's job — measuring how strong the best split is — *directly and deterministically*, with no search and no random seed.
- **$\#\{\mu_i > 0\}$.** The count of positive eigenvalues upper-bounds the number of separable communities (Fasino & Tudisco). It is a community count with no algorithm in it, and so a direct replacement for Louvain's unstable $k(t)$.
- **μ_N, the most negative.** Its eigenvector picks out the strongest *anti*-community structure: groups that avoid one another more than chance. In a market that would be a genuine split — defensives against cyclicals, or a flight to safety.
- **$\sum_{\mu_i > 0} \mu_i$.** Total "community mass" in the network.

None of these require Louvain, all are deterministic, and none of them appears anywhere in the finance literature. Figure 3 shows why the Laplacian cannot substitute for them, and Table 2 lists the three operators the probe actually computed.



Both axes are medians over all 6,315 days, at their own scales (**L**: one tick = 40; **B**: one tick = 10), aligned at zero.

Figure 3: The two spectra, drawn on the same networks. **L** is positive semi-definite, so its entire left half is unreachable; the only thing it can report about missing structure is how many pieces the graph has already fallen into, and once it has fallen into pieces λ_2 is identically zero and carries nothing. **B** straddles zero, and its negative half is a live signal about pairs of stocks that are avoiding each other.

Table 2: The three operators whose full spectrum the probe computes on each of the 6,315 daily networks. **C** is included because it is what exists *before* thresholding — the object the pipeline starts from.

	definition	what its spectrum reports
L	$\mathbf{D} - \mathbf{A}$	how well connected the network is, and where it would break. The multiplicity of the zero eigenvalue is the number of disconnected pieces
B	$\mathbf{A} - \mathbf{d}\mathbf{d}^\top/2m$	community structure (positive end) and anti-community structure (negative end)
C	the raw correlation matrix, *before* thresholding	how much everything is moving together; the leading eigenvalue is the market mode

4 One ruler for every candidate

Before any of these can be compared, they have to be put on a common scale. This section is the scoring apparatus; it is Silva's replication's own, unchanged.

Levels cannot be scored. λ_2 drifts over 25 years; "$\lambda_2 = 3.1$" means nothing on its own. What matters is whether today is unusual *relative to recent normal*. So every series is converted to a trailing 250-day z-score,

$$z(t) = \frac{x(t) - \text{mean}\left(x(t-250), \ldots, x(t-1)\right)}{\text{sd}\left(x(t-250), \ldots, x(t-1)\right)},$$

where $z = 0$ means today is typical and $z = 3$ means three standard deviations from the last year's normal. The window uses *only days strictly before today*, which is what makes this an honest detector rather than a description.

The crisis windows were fixed in advance. Seventeen events — Black Monday, Friday the 13th, Black Wednesday, October 1997, Russia/LTCM, Boo.com, 9/11, China 2007, Dubai 2009, and eight diffuse regimes — were written down in Phase 0 of the replication, before any network existed, so no detector can have been tuned to them. Nine are tagged *sharp*, and those are what everything below is scored against: 2.68% of days, about 169 of the 6,315.

Table 3: The same numbers with their definitions. "Operator" is which matrix the statistic is read from. All are scored identically; Q and τ reproduce the replication's stored values to ten decimal places.

Signal	Operator	What it measures	AUC
spectral entropy of $\mathbf{C}$	$\mathbf{C}$	how spread out the correlation structure is	0.822
$\lambda_1(\mathbf{C})/N$	$\mathbf{C}$	share of variance in the top common factor	0.812
$\tau(t)$	—	the cut-off needed that day to keep 10% of pairs	0.807
$\mu_N(\mathbf{B})$	$\mathbf{B}$	$\star$ strongest *anti*-community	**0.760**
largest-piece share	$\mathbf{L}$	fraction of stocks in the biggest component	0.757
# components	$\mathbf{L}$	zero eigenvalues of $\mathbf{L}$: how many pieces	0.756
$\sum_{\mu_i>0}\mu_i$	$\mathbf{B}$	$\star$ total community mass	**0.750**
$\lambda_{\max}(\mathbf{L})$	$\mathbf{L}$	the densest node's local connectivity	0.731
$\#\{\mu_i > 0\}$	$\mathbf{B}$	$\star$ community count, no algorithm	**0.722**
$\lambda_2(\mathbf{L})$, largest piece	$\mathbf{L}$	Fiedler value with fragmentation removed	0.652
$\mu_1 - \mu_2$	$\mathbf{B}$	modularity spectral gap	0.636
# modes above the noise band	$\mathbf{C}$	how many groups are even resolvable	0.619
$\lambda_2(\mathbf{L})$, whole graph	$\mathbf{L}$	Fiedler value — **Paper 1's indicator**	0.617
$\mu_1(\mathbf{B})$	$\mathbf{B}$	leading modularity eigenvalue	0.607
$Q(t)$	partition	Louvain modularity — **Paper 3's indicator**	**0.538**
$\max_k(\lambda_{k+1} - \lambda_k)$	$\mathbf{L}$	largest Laplacian gap (Kang et al.)	0.526

Accuracy is useless at that base rate, since a detector that says "no crisis" every day is 97.3% accurate and worthless. So the score is ROC-AUC:

What AUC means.
Rank all 6,315 days by the detector's z-score. Then **AUC is the probability that a randomly chosen crisis day ranks above a randomly chosen calm day.** 0.50 is a coin flip and carries no information; 1.00 is perfect; 0.75 means that picking one crisis day and one calm day at random, three times in four the crisis day scores higher.

Two things to declare about this ruler, both of which cut against the numbers rather than for them.

1. **Each statistic was tried in both directions** — "crisis = spike" and "crisis = drop" — and the better score kept. That flatters every entry in the table slightly, the new ones and the incumbents alike.
2. **λ_2's detector is undefined on some days.** On the 63% of days when the graph is disconnected λ_2 is exactly zero, and over long stretches of those the trailing standard deviation is zero too, so the z-score does not exist. This is a genuine property of the Fiedler value on these graphs rather than a scoring artifact, and it is part of why it scores as it does.

The point of the apparatus is that *every* statistic — incumbent and new — goes through the identical detector. As a control, the probe re-derives the replication's own stored scores for Q and τ and gets 0.5380684370 and 0.8071316467, matching to ten decimal places. The new numbers therefore sit on exactly the ruler the published ones do.

5 Discrepancy in the scalar: the reduction loses the signal

Silva's pipeline already built 6,315 daily networks. The probe walks through them and computes the full spectrum of $\mathbf{L}$, $\mathbf{B}$ and $\mathbf{C}$ for each one — three 360×360 eigendecompositions per day, about six minutes on one core — and scores every resulting statistic through the detector of §4. Figure 4 and Table 3 are the result.

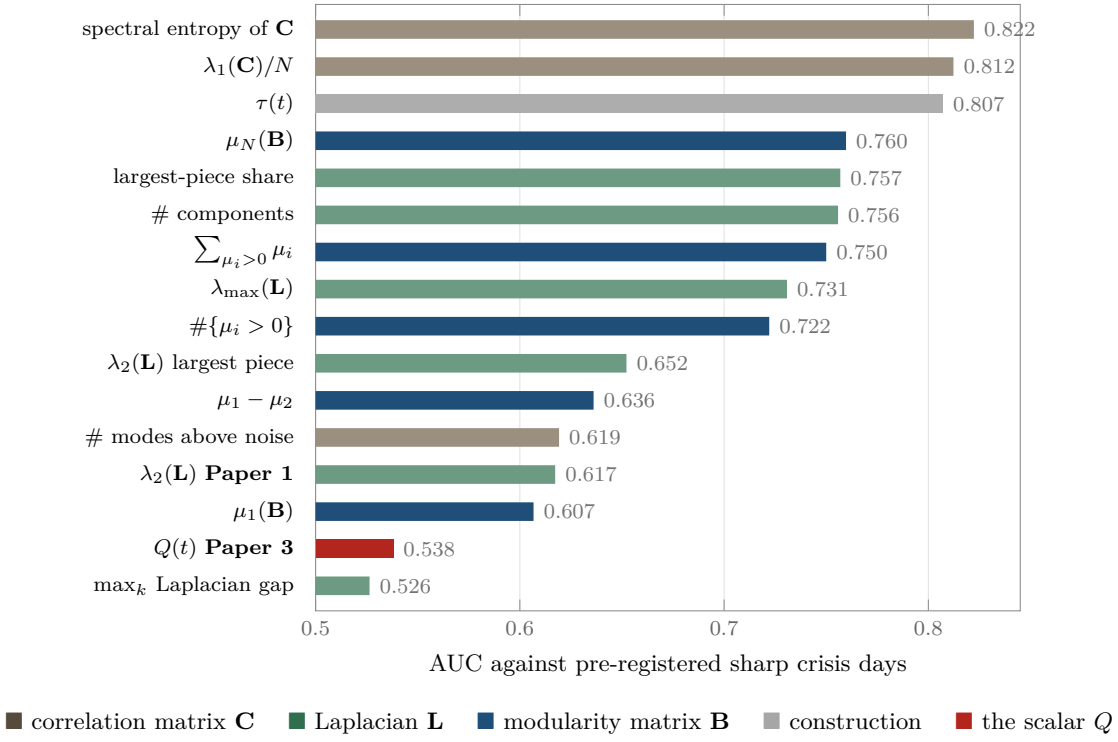


Figure 4: Every statistic, one ruler. Bars begin at 0.50, so bar length is information above a coin flip. $Q(t)$ — the quantity the paper is built on — is the shortest bar but one. Three readings of the matrix it is derived from, in dark blue, reach 0.722 to 0.760.

5.1 How to read $Q = 0.538$

Pick a random crash day and a random calm day. Q ranks the crash day higher 53.8% of the time. A coin manages 50%. Silva et al. build a 25-year narrative on this quantity, and against the crises that narrative is about, it is very nearly uninformative.

And it came from a matrix that scores 0.760**.** The three starred rows — $\mu_N(\mathbf{B})$, $\sum_{\mu>0}\mu$, and $\#\{\mu > 0\}$ — are statistics of the *same matrix* $\mathbf{B}$, on the *same networks*, under the *same scoring*, and they reach 0.722 to 0.760. Nothing about the data changed. The only thing that changed is that the matrix was read directly instead of being pushed through Louvain and collapsed to a scalar.

That is the key discrepancy. Three details fill it in.

The strongest modularity signal is the negative end. μ_N — anti-communities, groups pulling apart — beats every positive-side statistic of $\mathbf{B}$. That is precisely the thing thresholding is structurally incapable of representing (Figure 3), and it is unexamined in finance.

μ_1 **measures what Louvain is trying to measure, without the instability.** In levels $\mu_1(\mathbf{B})$ correlates 0.761 with Q, so it is largely tracking the same underlying quantity — but deterministically. Louvain is not: its seed-to-seed spread on a single day is 2.09× the median day-to-day move in Q. Run it twice on identical data and the disagreement is twice the size of the signal being studied. Even as a straight drop-in replacement that is worth something.

What this does *not* say. Paper 1's λ_2 scores 0.617 here, and that is not a refutation of Paper 1. Cardoso & Palomar learn a regularised Laplacian on seven stocks; this is a thresholded 360-node correlation network — a different object entirely. The claim is narrower and should stay narrow: *on thresholded correlation networks of this kind*, the Fiedler value is a weak detector. The same caution applies to Kang et al.'s largest-gap statistic at 0.526.

6 Three mechanisms behind the empty scalar

A discrepancy is worth more when its cause can be named. Three separate things are going wrong at the Louvain step, and they are independent of one another.

6.1 Louvain disagrees with itself

Louvain is a greedy randomised search, so it returns a different partition on every run. The replication measured how different. Normalised mutual information between two seeds on the *same day* is 0.543; between consecutive days at a *fixed* seed it is 0.463. In other words, two runs on identical data agree only 0.079 more than runs on genuinely different data. Expressed in the units of the series itself, the seed-to-seed spread on one day is **2.09**× the median day-to-day move in Q.

The consequence for a time series is direct: a meaningful fraction of the day-to-day variation in $Q(t)$ — the variation the paper's 25-year narrative is about — is the random number generator.

6.2 Louvain always finds communities, including in noise

Louvain has no way to answer "there is nothing here". It always returns a partition, and that partition always has a positive Q. The replication ran the control the paper did not: push i.i.d. Gaussian returns — containing no communities whatsoever — through the entire pipeline unchanged.

The noise control.
Pure noise gives $Q = 0.2246 \pm 0.0067$, with a median of 7 communities.
The real market gives $Q = 0.2215 \pm 0.0413$.

Structureless data comes out marginally more modular than the actual stock market.

The mechanism is not mysterious. A correlation matrix is positive semi-definite, so a high ρ_{ij} and a high ρ_{ik} force ρ_{jk} up. Thresholding one therefore produces a graph that is *transitive by construction* — full of triangles — and transitive graphs are modular. Measured on the noise networks, transitivity is 0.219. A degree-preserving null model cannot control for this, because rewiring destroys the geometry along with the structure.

The right comparison is therefore each series against *its own* null. The market's $Q = 0.2215$ sits +0.1150 above its degree-sequence null of 0.1065. A second noise run, scored the same way, gives $Q = 0.2253$ against a null of 0.1332 — an excess of +0.0921.[1] So 80% **of the market's apparent excess modularity is reproduced by data with no communities in it.**

6.3 At this window length, there is very little to detect

The previous subsection shows it happens. Random-matrix theory explains why it must.

Each day, 64,620 pairwise correlations are estimated from 30 days of data on 360 stocks — 10,800 numbers. That is six times more output than input, and it has an immediate arithmetic consequence: a correlation matrix built from 30 observations has **rank at most** 29, so at least 331 of its 360 eigenvalues are exactly zero, forced by counting rather than by anything about markets.

Marchenko–Pastur says how large the remaining ones get from estimation error alone. With $N/\Delta t = 360/30 = 12$, the noise band is

$$[(1 - \sqrt{12})^2,\ (1 + \sqrt{12})^2] \ = \ [\,6.07,\ 19.93\,].$$

[1] The two noise figures come from different runs and are not interchangeable: 0.2246 ± 0.0067 is the full-length Phase-3 level comparison quoted above, while 0.2253 is the 101-day Phase-4 run that also carries its own matched null, which is what the excess calculation needs. Both are recorded in `results/phase4_nulls.json` and `PITFALLS.md`.

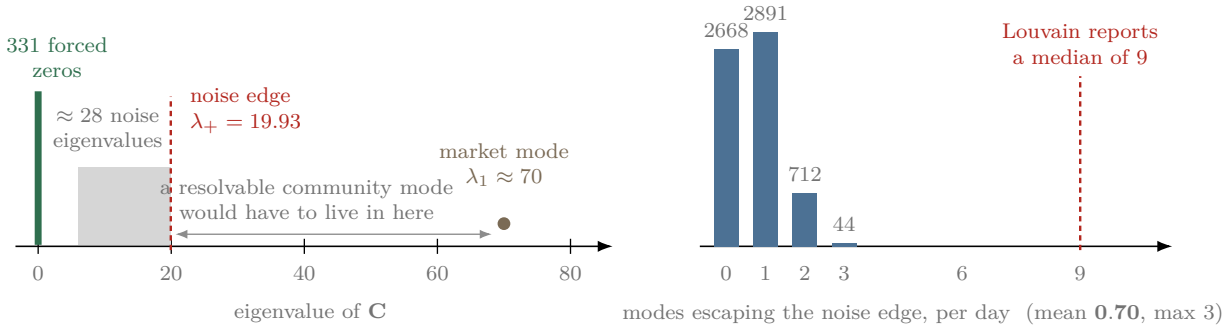


Figure 5: Left: where the eigenvalues of one day's correlation matrix actually sit. **Right:** how many of them clear the noise edge, counted over all 6,315 days, against the number of communities Louvain reports. There is on average less than one statistically resolvable group per day beyond the market itself, and none at all on 2,668 days.

Any eigenvalue below 19.93 **is indistinguishable from what pure noise produces.** It is not evidence of structure. Counting, for each day, the eigenvalues that escape above 19.93 — excluding the largest, which is the market mode — gives the right-hand panel of Figure 5.

The result:

> **Detectable structure per day.**
>
> Mean count **0.70**. Maximum 3. Exactly zero on **2,668** of 6,315 days.
>
> Louvain, on the same data, reports a **median of nine** communities every day, and the paper narrates their 25-year dynamics.

Most of those nine are not detectable structure; they are the algorithm partitioning noise. And this does not indict Silva et al. alone — $\Delta t = 30$ with a few hundred stocks is standard across this entire literature.

7 Discrepancy in the duality: two indicators that should be one number

The prompt presupposes that comparing the Laplacian and the modularity matrix is a meaningful thing to do. It is worth checking that presupposition, because there is a theorem that appears to destroy it.

The identity. On a connected graph in which *every* node has the same degree d, it is exactly true that

$$\lambda_2(\mathbf{L}) \; + \; \mu_1(\mathbf{B}) \; = \; d.$$

Both operators are built on top of the same adjacency matrix, and on a regular graph both reduce to reading the adjacency spectrum — one from the bottom, one from the top. **They are the same number, backwards.**

If that held on these networks, then "compare Paper 1's indicator with Paper 3's" would be a null question. There would be one indicator with two names, and the study would be measuring nothing.

Why it ought to nearly hold here. Silva's fixed-density construction pins the average degree at exactly 35.90 every single day — not approximately, by construction (§2). These graphs are about as close to regular as any real financial network one could build.

What was measured. Figure 6 and Table 4.

Read the residual first. $\lambda_2+\mu_1$ should equal 35.90 and instead comes out at 26.41: **the identity misses by** 26% **of the mean degree.** And the ±4.63 matters as much as the 9.49 — the gap is

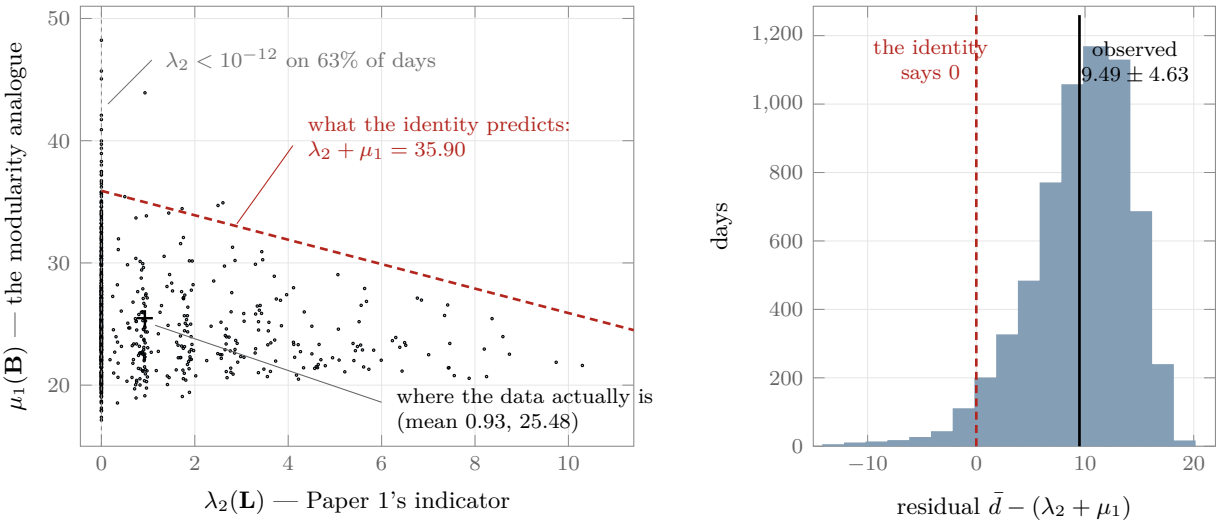


Figure 6: Left: the two indicators, one point per eight days. If the identity held, every point would sit on the red line. **Right:** the distribution of the shortfall. It is centred on 9.49, not zero, and it is *wide* — so it is not a fixed offset that could be subtracted away. The vertical stripe at $\lambda_2 \approx 0$ is the fragmentation problem in one glance: on 63% of days the graph is already in pieces, the Fiedler value is identically zero, and it cannot vary with μ_1 at all.

Table 4: The identity, tested three ways. The last two rows remove the two obvious confounds — days when the graph is in pieces, and the pieces themselves — and neither rescues it.

quantity	value	what it should be
average degree $\bar{d}$	35.900	exact, every day, by construction
observed $\lambda_2 + \mu_1$	26.411	35.900
residual $\bar{d} - (\lambda_2 + \mu_1)$, all days	$\mathbf{9.489 \pm 4.627}$	0
restricted to the 2,342 one-piece days	9.210 ± 3.595	0
on the largest component only	11.109 ± 5.314	0
$\mathrm{corr}(\lambda_2, \mu_1)$, all days	$\mathbf{-0.187}$	-1
one-piece days	-0.119	-1
largest component	-0.169	-1

not a fixed offset one could subtract away, it moves by a third of its own size from day to day. It is a live quantity.

The correlation is the cleaner statement. If $\lambda_2 + \mu_1 = \bar{d}$ with $\bar{d}$ fixed, then whenever λ_2 rises μ_1 must fall by exactly as much: they would be perfectly anti-correlated, $r = -1$. Two variables with no relationship at all give $r = 0$. Observed: $\mathbf{r = -0.187}$. That is far closer to unrelated than to identical.

Dual in theory, unrelated in practice.
The two indicators the field has been using are *provably the same number* in the idealised case and *empirically almost independent* on real market networks.

This is why the discrepancy is worth something rather than merely being a defect. It rescues the comparison question. It stops being the weak "which indicator wins?" and becomes a sharp one:

What is that 9.49 made of, and does it carry market-stress information?

There are three candidate constituents, and they are separable:

1. **Degree heterogeneity.** The identity assumes every node has degree d. Only the *average* is

pinned at 35.90; in reality some stocks have five edges and some have a hundred and fifty.

2. **Fragmentation.** The graph breaks into disconnected pieces on 63% of days (and on 99.3% of days after 2002). Whenever it does, λ_2 is identically zero — it cannot vary with anything. That is the vertical stripe in Figure 6, and it is also why restricting to the largest component makes the residual *worse*, not better.
3. **The market mode.** The common factor that dominates $\mathbf{C}$ (§8) enters both operators, but not symmetrically.

Establishing which of the three carries the stress information is a structural result about what the field's two incumbent indicators actually measure. It does not depend on winning a horse race. That is direction D1.

8 What weakens all of this

Three things cut against the results above. They are put here rather than in a footnote because they are the first things a referee will raise, and because the third one is currently unresolved.

8.1 The strongest detectors are one signal, and it was found in 1999

On any given day the single biggest pattern in stock returns is that everything moves together. In $\mathbf{C}$ this shows up as one dominant eigenvalue λ_1, vastly larger than the rest; in a crash, correlations go towards one across the board and λ_1 balloons. That is the mechanism behind "diversification fails exactly when you need it".

This is old, established finance. $\lambda_1(\mathbf{C})/N$ is precisely the **absorption ratio** of Kritzman et al. (2011), and its random-matrix precursors are Laloux et al. and Plerou et al., 1999.

Now look again at the top of Figure 4: spectral entropy 0.822, absorption ratio 0.812, $\tau(t)$ 0.807. Those three are **pairwise correlated above** 0.99 **in levels**. A correlation of 0.99 means they are the same series with different labels. And the component count (0.756) correlates 0.92 with τ, so really it is four.

> **The top of the table is not four discoveries.**
> It is one discovery, and it was made in 1999.

8.2 Out of sample, nothing yet beats that baseline

The real question is therefore whether the modularity spectrum adds anything *on top of* the market mode. Testing that honestly means fitting on one period and evaluating on data never seen: fit before 2002, evaluate after (Table 5, Figure 7).

Note that the ranking inverts completely. Adding Q looks like the second-best idea in-sample (0.831) and is among the worst out of sample (0.699).

The mitigating detail, and why it is not a defence. Twenty-seven crisis events in the test set is a very small number; AUC on 27 events is noisy enough that none of this is statistically conclusive in *either* direction. But "our evidence is too weak to settle it" is not a rebuttal — it is the same statement. As things stand there is *no evidence* that the modularity spectrum adds anything to a 1999 baseline out of sample.

8.3 How much of μ_N's score is borrowed

$\mu_N(\mathbf{B})$ scored 0.760, the best new number in the probe. But on the scored detector series it correlates 0.680 with the market mode. So a substantial part of that 0.760 is borrowed from a

Table 5: Logistic detector, fitted on pre-2002 and evaluated on post-2002 (27 crisis days in the test period). Every addition helps in-sample and hurts out of sample.

model	in-sample	out-of-sample
market mode alone	0.812	**0.730**
+ $\mu_1(\mathbf{B})$	0.822	0.629
+ $\mu_N(\mathbf{B})$	0.810	0.719
+ $Q(t)$	0.831	0.699
+ $\lambda_2(\mathbf{L})$ on the largest component	0.810	0.700
market mode + the whole modularity spectrum	0.823	0.640
modularity spectrum alone	0.806	0.685
$Q(t)$ alone	0.538	0.667

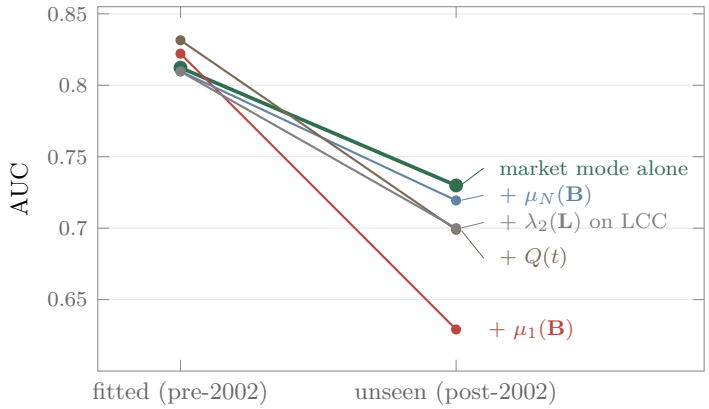


Figure 7: The same table as slopes. Every line falls, which is expected — but the plain market mode falls least and finishes highest, and adding μ_1 produces both the largest in-sample gain and the largest out-of-sample loss. That combination is the textbook fingerprint of overfitting: the extra variable is not learning a market mechanism, it is memorising 1987–2002.

signal that already existed.

A note on which series these correlations are measured on.
The 0.680 above, and the 0.025 below, are correlations between the *signed, 250-day z-scored detector series* — i.e. between the things actually being scored in Figure 4, which is the comparison that matters. In raw levels the same two pairs come out at -0.819 and $+0.294$. Both bases are legitimate; they answer different questions, and the earlier write-ups did not say which was meant.

There is one genuine bright spot. $\mu_1(\mathbf{B})$ correlates only **0.025** with the market mode — essentially orthogonal — while correlating 0.761 with Q in levels. So it really is a new axis, measuring what Louvain is trying to measure without duplicating the incumbent. It simply did not convert that independence into out-of-sample gain, and that tension is unresolved.

8.4 Two smaller cautions

The Laplacian numbers are on Silva's graphs, not Paper 1's — restated here because it is easy to lose. Extending the comparison to a learned Laplacian is itself part of D1.

Everything here is a one-pass probe. These runs were designed to establish which direction is worth months of work, not to be results in a paper.

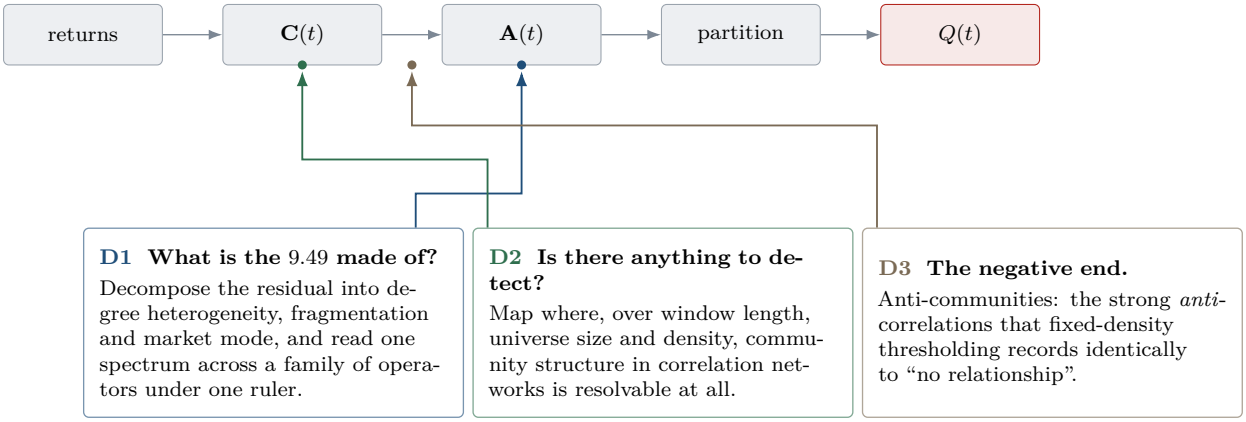


Figure 8: Where each direction attacks. D2 asks whether **C** contains resolvable structure in the first place; D3 concerns what the thresholding arrow throws away; D1 concerns which operator the resulting graph should be read through. The boxes are drawn identically on purpose.

9 Three possible directions

Superseded, 8 September 2026. Directions D1 and D2 below rest on the identity $\lambda_2(L) + \mu_1(B) = d$, which holds only on a regular graph. These graphs are not regular, so it is not an identity here. The later report, `reports/2026-09-08_modularity_equivalence.tex`, replaces that framing with a proof: after degree normalisation B and L are the same operator ($B_n = I - L_{\text{sym}} - uu^\top$, shared eigenvectors, eigenvalues paired as $\mu = 1 - \lambda$, verified to 1.8×10^{-15} on all 243 windows), so no normalised-B indicator can carry anything new, and the unnormalised shortcut $\bar{d} - \mu_1(B)$ is unsound. The measurements in this section stand; the spine built on them does not. D3 is unaffected.

Three directions follow from the above. They are set out below in label order, in an identical frame, at the same length. **No ordering or preference is implied, and they are not combined into a recommendation** — Table 6 is deliberately symmetric, so that nothing in its layout favours one direction over another.

Figure 8 shows where each one attacks the pipeline of Figure 1, which is the clearest way to see that they are genuinely different questions rather than three versions of one.

One factual note that belongs with D3 rather than with any ranking: the discarded anti-correlation series *collapses to near zero from roughly 2003 and never recovers*, which sits directly beside the unexplained 2002 break that Silva et al. report and list as future work. Whether the two are connected is untested.

A Where every number comes from

Nothing in this document is retyped from a figure. Each row below names the script that produces the value and the file it is stored in; all paths are relative to the repository root. The probe imports Silva's replication's own `utils/crisis_dates.py`, `net/construct.py` and scoring conventions unchanged.

Table 6: The three directions in a common frame. Rows are identical questions asked of each; nothing in the table ranks them.

	D1 the operator comparison	**D2** detectability	**D3** the negative end
The question	What is the 9.49 residual made of, and which part of it carries stress information?	Over what region of (window, universe, density) does community structure in correlation networks mean anything?	Do anti-communities carry crisis information that communities do not?
Evidence so far	$\lambda_2 + \mu_1 = 26.41$ against 35.90; $r = -0.187$ against -1 (§7).	Mean 0.70 resolvable modes per day against Louvain's median of nine; noise $Q = 0.2246$ vs market 0.2215 (§6).	$\mu_N(\mathbf{B})$ at 0.760, the highest-scoring new statistic (§5); 4,246,798 discarded strong-anticorrelation pair-days.
What exists already	The full spectra for all 6,315 days, and the residual decomposition scaffolding, in `probe/duality.py`.	The Marchenko–Pastur edge and per-day escaping-mode counts, in `probe/spectral_probe.py`.	The per-day μ_N series, and the audit's count of discarded pair-days.
What would be built	One spectrum over **A**, **L**, normalised **L**, **B**, normalised **B**, and MacMahon–Garlaschelli's threshold-free correlation modularity, under the ruler of §4.	A grid over Δt, N and density, benchmarked against both the MP edge and the Nadakuditi–Newman detectability threshold; plus a deterministic Louvain-free community count.	The anti-correlation series reconstructed and scored; μ_N re-scored with the market mode projected out.
Currently untested	Whether the decomposition is stable across network constructions, not just Silva's.	Whether any configuration used in the literature falls inside the resolvable region.	Whether μ_N survives removing the market mode — it borrows 0.680 from it on the scored series (§8). A two-day check.
What could go wrong	The gap may turn out to be dominated by fragmentation, i.e. an artifact of thresholding rather than a market property.	If essentially no configuration in the literature is inside the region, the contribution is purely negative — publishable, but harder to place.	If μ_N does not survive market-mode removal, the result folds back into D1 as one component of the residual.

Numbers	**Produced by**	**Stored in**
All 16 AUCs (Table 3)	`probe/spectral_probe.py`	`probe/spectral_probe_auc.json`
λ_2 on the largest component	`probe/duality.py`	`probe/duality_auc.json`
Per-day spectra, $6{,}315 \times 16$	`probe/spectral_probe.py`	`probe/spectral_probe.parquet`
Residual, $\mathrm{corr}(\lambda_2, \mu_1)$ (Table 4)	`probe/duality.py`	the parquet above
Escaping-mode counts (Figure 5)	`probe/spectral_probe.py`	the parquet above
In/out-of-sample AUCs (Table 5)	`probe/incremental.py`	`probe/incremental_auc.json`
Louvain stability, noise-control Q	Silva replication, phases 3–8	`results/phase8_extensions.json`
Discarded anticorrelated pair-days	Silva replication, phase 2	`PITFALLS.md` (A8)

All of it is re-checked by `reports/probe/check_numbers.py`, which recomputes every value quoted here from those files and fails loudly on any mismatch.

Reproducing. The probe needs the replication's derived artifacts (`data/edges.npy`, `data/returns.npy`, `results/series/*.parquet`), which are gitignored as rebuildable; regenerate them with `python run_phases.py 1 3` inside `paper3_market_modularity/` first. Silva's pipeline needs `python-igraph`.

Outside references used above. Kritzman et al. 2011 (absorption ratio, the baseline to beat); Laloux et al. and Plerou et al. 1999 (RMT precursors); Newman 2006 (the modularity matrix); Fasino & Tudisco 2014/2016/2017 (modularity spectrum, community counts, anti-communities); Nadakuditi & Newman 2012 (detectability); MacMahon & Garlaschelli 2015 (the correlation-specific null Silva et al. needed and did not use); Masuda et al., *Physics Reports* 2025 (the "beyond thresholding" review); Kang et al. 2025 (largest Laplacian gap). Papers replicated: arXiv:2005.09958 (Cardoso & Palomar), arXiv:1706.03161 (TICC), arXiv:1501.05040 (Silva et al.). These are a working set of pointers rather than a systematic review: most are recorded from abstracts and stated main results rather than full readings, and none is relied on for any number in this document. Two Laplacian-energy papers previously listed here (*Eur. J. Finance* 2023; *Int. Rev. Econ. Finance* 2025) are removed pending verification of their author lists.